Debamita Ghosh

⌂ Department of Electrical and Computer Engineering (ECE), University of Central Florida
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Research Interests

Theoretical foundations of reliable sequential decision-making under uncertainty, with an emphasis on online learning, structured multi-armed bandits, distributionally robust reinforcement learning, multi-agent reinforcement learning, and robust interactive imitation learning. My research focuses on developing statistically efficient algorithms with finite-sample guarantees that remain reliable under limited feedback, model uncertainty, distribution shifts, and strategic multi-agent interactions, with applications to next-generation wireless networks and autonomous intelligent systems.

Current Position

Postdoctoral Research Fellow

University of Central Florida (UCF)
Department of Electrical and Computer Engineering (ECE)
Duration: March 2025 – Present
Supervisors: Prof. George K. Atia, Dept. of ECE, UCF
Prof. Yue Wang, Dept. of ECE, UCF

Education

PhD

IITB-Monash Research Academy, Indian Institute of Technology Bombay, India
Department of Industrial Engineering and Operations Research (IEOR)
Duration: July 2019 – July 2025
Supervisors: Prof. Manjesh Kr. Hanawal, Dept. of IEOR, IIT Bombay, India
Prof. Nikola Zlatanov, Faculty of Computer and Engineering Sciences, Innopolis University, Russia.
Thesis Title: Online Algorithms for Efficient Beamforming in Next-Generation Wireless Networks

Masters in Statistics

Indian Institute of Technology Kanpur, India
Department of Mathematics & Statistics
Duration: July 2010 – June 2012

Bachelors in Statistics

St. Xavier's College (Autonomous), Kolkata, India
Department of Statistics
Duration: July 2006 – June 2010

Publications

Peer-reviewed Journals

1. Ghosh, D., Hanawal, M.K. and Zlatanov, N. (2024). Fixed Budget Best Arm Identification in Unimodal Bandits. *Transactions on Machine Learning Research (TMLR).*
2. Ghosh, D., Hanawal, M.K. and Zlatanov, N. (2024). UB3: Fixed Budget Best Beam Identification in mmWave Massive MISO via Pure Exploration Unimodal Bandits. *IEEE Transactions on Wireless Communications,* doi: 10.1109/TWC.2024.3395129.
3. Ghosh, D., Hanawal, M.K. and Zlatanov, N. (2023). Learning Optimal Phase-Shifts of Holographic Metasurface Transceivers. *IEEE Transactions on Wireless Communications.*
4. Ghosh, D., Hanawal, M.K. and Zlatanov, N. (2021). Learning to Optimize Energy Efficiency in Energy Harvesting Wireless Sensor Networks. *IEEE Wireless Communications Letters,* 10(6), 1153–1157.

Conferences

1. Ghosh, D., Atia, G.K. and Wang, Y. (2026). Online Robust Reinforcement Learning with General Function Approximation. *Proceedings of the 43rd International Conference on Machine Learning (ICML).* arXiv:2512.18957.
2. Ahmed, S., Ghosh, D., Tang, D., Wang, Y., Velasquez, A., & Wang, Y. (2026). Robust Nash Alignment under Preference Uncertainty. *Advances in Neural Information Processing System (NeurIPS).* arXiv preprint arXiv:2610.00715.
3. Farhat, Z.U., Ghosh, D., Atia, G. and Wang, Y. (2026). Sample-efficient distributionally robust multi-agent reinforcement learning via online interaction. In *International Conference on Learning Representations (ICLR)*, Vol. 2026, pp. 151781-151838.
4. Ghosh, D., Atia, G.K. and Wang, Y. (2026). ORVIT: Near-Optimal Online Distributionally Robust Reinforcement Learning. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI),* Vol. 40, No. 25, pp. 21278–21286.
5. Ghosh, D., Hanawal, M.K. and Zlatanov, N. (2026). MBT: Multi-Beam Strategy for Best Transmit Beamforming in mmWave Communications. *2026 18th International Conference on COMmunication Systems and NETworks (COMSNETS),* pp. 147–155. IEEE.
6. Ghosh, D., Hanawal, M.K. and Zlatanov, N. (2025). Fixed Budget Bayesian Best Arm Identification in Unimodal Bandits. *2025 17th International Conference on COMmunication Systems and NETworks (COMSNETS),* pp. 748–756. IEEE.
7. Ghosh, D., Hanawal, M.K. and Zlatanov, N. (2024). HoloBeam: Learning Optimal Beamforming in Far-Field Holographic Metasurface Transceivers. *IEEE INFOCOM 2024,* pp. 301–310. IEEE.
8. Ghosh, D. and Hanawal, M.K. (2023). Fixed Budget Beam Identification in Millimeter Wave Systems via Unimodal Bandits. *2023 15th International Conference on COMmunication Systems & NETworkS (COMSNETS),* pp. 486–488. IEEE.
9. Ghosh, D., Verma, A. and Hanawal, M.K. (2020). Learning and Fairness in Energy Harvesting: A Maximin Multi-Armed Bandits Approach. *2020 International Conference on Signal Processing and Communications (SPCOM),* pp. 1–5. IEEE.

Academic Achievements

Naik and Rastogi Award 2026 — **Best Ph.D. Thesis Award**

Recipient of the Naik and Rastogi Award for Excellence in Ph.D. Research for the year 2024-2026 (Best Thesis Award of IIT Bombay) for doctoral research.

Coromandel Int. Ltd. 2025 — **Best Ph.D. Thesis Award**

Recipient of the Coromandel International Ltd. 2025 Best Ph.D. Thesis Award (Best Thesis Award of IITB-Monash Research Academy, IIT Bombay) for doctoral research.

INFOCOM CTG 2024

INFOCOM 2024 Conference Travel Grant (CTG)

Conference Travel Grant from INFOCOM association and IEEE Communications Society (ComSoc) awarded to selected student candidates who have an accepted paper as a first author in INFOCOM main conference.

COMSNETS 2023

Best Graduate Forum Presentation (Runner Up)

Received Best Graduate Forum Presentation (Runner Up) for "Fixed Budget Beam Identification in Millimeter Wave Systems via Unimodal Bandits" at COMSNETS 2023.

COMSNETS Association and LRN Travel Grant (2023)

Travel award from COMSNETS association and LRN foundation to selected student candidates who have an accepted paper as a first author in COMSNETS main conference.

Monash University

University Fellowship (April 2023 – August 2023)

Research training fellowship awarded to selected incoming graduate students in the Department of Electrical & Computer Systems Engineering, Monash University.

FATE-ML

LinkedIn and Microsoft Travel Grant (2020)

Travel Award from LinkedIn for FATE-ML workshop 2020 at IISc, Bengaluru, India.

IIT Bombay Fellowship

University Fellowship (July 2019 – July 2025)

Research training fellowship awarded to selected incoming graduate students in the Department of Industrial Engineering & Operations Research, IIT Bombay.

UGC Govt. of India

State Level Eligibility Test for Lectureship (2014)

Awarded by the University Grants Commission (UGC), Govt. of India, through State Eligibility Test (SET) for Lectureship (secured Rank 1), deeming eligible for College/University Lecturer in Mathematical Science.

Research & Teaching Experiences

July 2019 – July 2025

Indian Institute of Technology Bombay, Mumbai, India

Teaching Assistant (TA), Industrial Engineering and Operations Research (IEOR)

Courses Assisting:

- IE 617/613: Online Machine Learning and Bandit Algorithms
- IE 605: Engineering Statistics
- NPTEL – Bandit Algorithms, Introduction to Stochastic Processes, Engineering Statistics

Aug. 2016 – July 2019

Amity University, Kolkata, India

Assistant Professor (Grade-I), Amity Institute of Applied Science

- Teaching Responsibilities: Multivariate Analysis, Statistical Inference, Regression Analysis at undergraduate and postgraduate level.
- Administrative Responsibilities: Event Coordinator, Program Coordinator and Exam Coordinator of Applied Science, Amity University, Kolkata.

Sept. 2015 – Mar. 2016

Techno India Agartala, West Tripura, India

Assistant Professor (Contract), Dept. of Basic Sciences (Mathematics)

Aug. 2014 – Sept. 2015

Netaji Subhash Engineering College, Kolkata, India

Assistant Professor (Contract), Dept. of Basic Engineering Science (Mathematics)

Aug. 2012 – Mar. 2013	**Asutosh College, Kolkata, India** Guest Lecturer, Dept. of Statistics
May 2011 – July 2011	**Research Intern, Indian Statistical Institute (ISI), Kolkata, India** Project on "Forecasting of Purchasing Power of Indian Districts: An Analysis with Regression and Extrapolation" at ISEC, ISI Kolkata.

Conferences & Workshops

Talks

- *"Online Robust Reinforcement Learning with General Function Approximation",* ICML 2026.
- *"MBT: Multi-Beam Strategy for Best Transmit Beamforming in mmWave Communications",* COMSNETS 2026, Bengaluru, India.
- *"HoloBeam: Learning Optimal Beamforming in Far-Field Holographic Metasurface Transceivers",* IEEE INFOCOM 2024, Vancouver, Canada.
- *"Fixed Budget Beam Identification in Millimeter Wave Systems via Unimodal Bandits",* COMSNETS 2023, Bengaluru, India.
- *"Learning and Fairness in Energy Harvesting: A Maximin Multi-Armed Bandits Approach",* SPCOM 2020, Bengaluru, India.

Workshop

- Presented a talk on *"Analysis of data using R through examples*" at the workshop organized by the Bureau of Applied Economics & Statistics, West Bengal, India, June 2017.
- Attended workshop on *"Fairness, Accountability, Transparency and Ethics in Machine Learning (FATE-ML)",* organized by LinkedIn, Microsoft and IISc at IISc, Bengaluru, India, January 2020.
- Attended workshop on *"Data Analytics*", organized by Interdisciplinary Statistical Research Institute, Indian Statistical Institute, Kolkata, India, March 2019.

Professional Services

Journal Reviewer

- Journal of Machine Learning Research
- IEEE Transactions on Signal Processing
- IEEE Transactions on Networking
- IEEE Transactions on Machine Learning in Communications and Networking

Conference Reviewer

- International Conference on Machine Learning
- International Conference on Learning Representations
- Neural Information Processing Systems
- Uncertainty in Artificial Intelligence

Reference

Postdoctoral Advisors

Prof. George K. Atia
Dept. of ECE & CS
University of Central Florida, Florida
george.atia@ucf.edu

Prof. Yue Wang
Dept. of ECE & CS
University of Central Florida, Florida
yue.wang@ucf.edu

Ph.D. Advisors

Prof. Manjesh Kr. Hanawal
Dept. of IEOR
IIT Bombay, India
mhanawal@iitb.ac.in

Prof. Nikola Zlatanov
Faculty of Computer and Engineering Sciences
Innopolis University, Russia
n.zlatanov@innopolis.ru